

RUSTENBURG, South Africa

WMO: 682550

Lat: 25.6608S

Lon: 27.2322E

Elev: 1150

StdP: 88.25

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.2	3.8	-12.5	1.5	18.3	-10.1	1.8	18.4	5.4	15.7	4.5	15.7	1.1	230	0.297

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	34.4	17.2	33.1	17.3	31.8	17.1	22.6	28.2	22.0	27.3	21.3	26.4	2.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	18.2	24.8	20.5	17.5	24.2	19.8	16.7	23.6	73.2	28.1	70.5	27.4	68.0	26.7	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	5.1	4.3	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	23.9	23.8	22.3	19.0	15.9	13.3	12.8	15.9	19.9	22.1	22.8	23.7
	DBStd	4.72	2.41	2.15	2.38	2.48	2.48	1.95	1.91	2.81	3.10	2.95	2.76	2.62
	HDD10.0	6	0	0	0	0	1	2	2	1	0	0	0	0
	HDD18.3	541	1	0	3	20	81	152	172	84	21	5	3	1
	CDD10.0	3506	430	387	381	269	183	100	89	184	298	376	384	424
	CDD18.3	1000	173	153	125	39	4	0	0	9	69	122	138	167
	CDH23.3	8757	1409	1171	812	288	127	11	11	219	863	1229	1202	1414
	CDH26.7	3139	529	416	232	51	10	0	0	37	333	503	470	559
Wind	WSAvg	1.5	1.4	1.4	1.7	1.8	1.4	1.2	1.3	1.5	1.6	1.6	1.6	1.4
Precipitation	PrecAvg	614	109	86	83	44	18	6	2	4	14	56	80	105
	PrecMax	1078	231	318	282	126	112	42	16	34	55	145	158	183
	PrecMin	370	27	32	8	5	0	0	0	0	0	8	23	46
	PrecStd	165	50	64	58	34	26	10	4	7	17	35	36	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	34.8	33.1	30.7	28.1	24.8	24.8	29.6	33.6	35.2	35.4	36.1
		MCWB	18.7	18.1	19.0	18.9	13.1	11.7	10.9	13.7	14.5	15.6	16.3	17.6
	2%	DB	32.9	32.9	30.9	28.1	26.1	23.1	22.9	27.7	32.0	33.2	33.2	33.7
		MCWB	19.4	18.1	18.9	17.2	13.3	11.3	10.4	13.1	14.0	15.7	16.6	18.1
	5%	DB	31.3	31.0	29.2	26.6	24.8	21.8	21.2	25.9	30.6	31.5	31.4	31.6
		MCWB	19.2	18.7	18.8	16.9	13.4	11.1	10.3	12.1	13.5	15.6	16.9	18.5
	10%	DB	29.8	29.2	27.7	25.0	23.0	20.2	19.8	23.9	28.6	29.6	29.6	29.6
		MCWB	19.2	18.6	18.7	16.9	13.3	10.8	10.0	11.4	13.2	15.5	17.2	18.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	23.0	23.1	21.2	17.4	15.1	13.5	15.5	18.0	20.7	22.3	23.4
		MCDB	29.7	29.6	28.7	25.8	23.4	21.3	20.1	25.8	26.2	25.3	27.7	28.0
	2%	WB	22.4	21.8	21.6	20.0	16.1	13.4	12.5	14.5	16.4	19.5	21.4	22.5
		MCDB	28.8	26.8	27.2	25.2	21.7	19.3	18.9	24.8	26.4	25.1	26.4	27.2
	5%	WB	21.7	21.0	20.7	19.1	15.2	12.4	11.5	13.7	15.4	18.5	20.5	21.7
		MCDB	27.6	25.5	26.1	24.0	21.2	18.6	18.3	23.7	25.3	24.3	25.5	26.4
	10%	WB	21.0	20.2	19.9	18.2	14.2	11.6	10.6	12.7	14.6	17.6	19.5	20.9
		MCDB	26.1	24.8	25.2	22.7	20.3	18.0	17.8	22.1	24.7	24.0	24.9	25.5
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.6	10.6	12.3	15.1	15.4	15.3	15.9	15.9	13.6	11.8	11.0
		MCDBR	13.9	13.8	13.2	14.7	16.9	17.3	17.2	17.8	18.5	16.6	15.0	14.1
		MCWBR	4.2	4.6	4.6	5.6	6.8	7.6	7.4	7.3	6.4	5.2	4.5	4.2
	5% WB	MCDBR	12.0	10.2	11.0	11.0	13.9	14.7	14.4	16.2	15.2	12.0	11.1	10.8
		MCWBR	4.9	4.5	5.1	5.2	6.9	7.6	7.2	7.4	6.2	4.9	5.1	5.1
Clear-Sky Solar Irradiance	taub	0.395	0.399	0.387	0.343	0.303	0.287	0.288	0.347	0.390	0.421	0.376	0.382	
	taud	2.345	2.327	2.342	2.427	2.507	2.527	2.503	2.341	2.237	2.192	2.363	2.372	
	Ebn at Noon	949	929	911	907	906	903	915	889	893	898	960	963	
	Edh at Noon	135	134	126	105	88	82	87	113	137	152	132	132	
All-Sky Solar Radiation	RadAvg	6.68	6.31	5.61	4.80	4.42	4.00	4.34	5.14	6.12	6.60	6.82	6.86	
	RadStd	0.71	0.59	0.54	0.44	0.21	0.21	0.11	0.24	0.26	0.43	0.65	0.44	

Historical Trends

		Heating			Cooling			Degree-Days		
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	+0.66	+0.58	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page